

Course 5103C

5 Credits: 4

Program Elective

Source-grounded

Screen Printing

□ To guide the student to the basic concepts of the screen printing

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5103C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5103C.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Printing Technology
Course code	5103C
Course title	Screen Printing
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

19 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To guide the student to the basic concepts of the screen printing
- To provide the exposure to the essentials of screen printing and to learn how it influences the real printing world.
- To understand the stencil concept to the printing process.
- To develop idea on most exciting possibilities of screen printing in the upcoming era.

Course outcomes

CO	Official outcome	Study level
CO1	benefits, applications and various materials 15 Understanding required to achieve the screen printing process. Explain about substrates and inks used in screen	See official syllabus
CO2	printing and various ink drying methods and 14 Understanding related facilities. Explain about on contact & off contact printing,	See official syllabus
CO3	specialized areas, image transfer and registration 15 Understanding techniques. Describe about various printing press	See official syllabus
CO4	14 Understanding classifications & configurations of screen-printing.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2

Row	Extracted official mapping
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	and various materials required to achieve the screen printing process.	7	As specified
Module 2	drying methods and related facilities.	4	As specified
Module 3	transfer and registration.	4	As specified
Module 4	screen-printing.	4	As specified

MODULE 1

and various materials required to achieve the screen printing process.

This module is presented from the official syllabus scope for Screen Printing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and various materials required to achieve the screen printing process.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

History of Screening Printing, 1 Understanding Merits, Demerits and applications of screen

History of Screening Printing, 1 Understanding Merits, Demerits and applications of screen is an official topic in Module 1 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify History of Screening Printing, 1 Understanding Merits, Demerits and applications of screen using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, history of screening printing, 1 understanding merits, demerits and applications of screen should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What History of Screening Printing, 1 Understanding Merits, Demerits and applications of screen means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എറു History of Screening Printing, 1 Understanding Merits, Demerits and applications of screen എറുഎറുഎറുഎറുഎറുഎറുഎറുഎറു definition/meaning എറുഎറുഎറുഎറുഎറു. എറുഎറുഎറു എറുഎറുഎറുഎറു, steps, application എറുഎറുഎറുഎറുഎറു എറുഎറുഎറു. Technical terms English-എറു എറുഎറു എറുഎറുഎറുഎറുഎറു.

1 Understanding printing. Screen printing materials- Screen, Frame,

1 Understanding printing. Screen printing materials- Screen, Frame, is an official topic in Module 1 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding printing. Screen printing materials- Screen, Frame, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding printing. screen printing materials- screen, frame, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding printing. Screen printing materials- Screen, Frame, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-1 Understanding printing. Screen printing materials- Screen, Frame, definition/meaning. steps, application. Technical terms English-1

3 Understanding Squeegee and Stencil. Screens – types, selection, stretching,

3 Understanding Squeegee and Stencil. Screens – types, selection, stretching, is an official topic in Module 1 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Squeegee and Stencil. Screens – types, selection, stretching, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding squeegee and stencil. screens – types, selection, stretching, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Squeegee and Stencil. Screens – types, selection, stretching, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-3 Understanding Squeegee and Stencil. Screens - types, selection, stretching, സ്ക്രീൻ സ്ക്രീൻ definition/meaning സ്ക്രീൻ സ്ക്രീൻ. സ്ക്രീൻ സ്ക്രീൻ സ്ക്രീൻ, steps, application സ്ക്രീൻ സ്ക്രീൻ സ്ക്രീൻ. Technical terms English- സ്ക്രീൻ സ്ക്രീൻ സ്ക്രീൻ.

3 Understanding preparation & reclamation. Trouble shooting clogged screens, Care and

3 Understanding preparation & reclamation. Trouble shooting clogged screens, Care and is an official topic in Module 1 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding preparation & reclamation. Trouble shooting clogged screens, Care and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding preparation & reclamation. trouble shooting clogged screens, care and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding preparation & reclamation. Trouble shooting clogged screens, Care and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding preparation & reclamation. Trouble shooting clogged screens, Care and $\text{ശരിയായി വളർത്തുന്നതിനുള്ള ശ്രമങ്ങൾ}$ definition/meaning $\text{എന്താണ് മണ്ണിലെ പദാർത്ഥം}$. $\text{മണ്ണിലെ പദാർത്ഥം എന്താണ്, steps, application}$ $\text{മണ്ണിലെ പദാർത്ഥം എന്താണ്. Technical terms English-ൽ എഴുതുക}$ മണ്ണിലെ പദാർത്ഥം .

1 Understanding storage of screens.

1 Understanding storage of screens. is an official topic in Module 1 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding storage of screens. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding storage of screens. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding storage of screens. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding storage of screens.

മലയാളത്തിൽ നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

Squeegee considerations.

Squeegee considerations. is an official topic in Module 1 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Squeegee considerations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, squeegee considerations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Squeegee considerations. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Squeegee considerations. Squeegee definition/meaning. Squeegee steps, application. Technical terms English- Malayalam.

Stencils-classifications and preparation. 3 Understanding Content: History of Screening Printing, Advantages, Disadvantages and Applications of screen printing. Screen printing materials: Screen, Frame, Squeegee and Stencil. Screens - multifilament, mono filaments, Selecting mesh material, stretching screen fabric to frame, screen preparation, screen reclamation - Trouble shooting clogged screens. Care and storage of screens. The squeegee, Squeegee considerations, squeegee preparation, hardness categories of squeegee blades, Variety of blade shape and application. Stencils-types, preparation methods of - knife cut stencils, photo stencils - Indirect stencil systems, Direct photo stencil systems, capillary systems, Direct/Indirect photo stencil systems. Explain about substrates and inks used in screen printing and various ink

Stencils-classifications and preparation. 3 Understanding Content: History of Screening Printing, Advantages, Disadvantages and Applications of screen printing. Screen printing materials: Screen, Frame, Squeegee and Stencil. Screens - multifilament, mono filaments, Selecting mesh material, stretching screen fabric to frame, screen preparation, screen reclamation - Trouble shooting clogged screens. Care and storage of screens. The squeegee, Squeegee considerations, squeegee preparation, hardness categories of squeegee blades, Variety of blade shape and application. Stencils-types, preparation methods of - knife cut stencils, photo stencils - Indirect stencil systems, Direct photo stencil systems, capillary systems, Direct/Indirect photo stencil systems. Explain about substrates and inks used in screen printing and various ink is an official topic in Module 1 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Stencils-classifications and preparation. 3 Understanding Content: History of Screening Printing, Advantages, Disadvantages and Applications of screen printing. Screen printing materials: Screen, Frame, Squeegee and Stencil. Screens - multifilament, mono filaments, Selecting mesh material, stretching screen fabric to frame, screen preparation, screen reclamation - Trouble shooting clogged screens. Care and storage of screens. The squeegee, Squeegee considerations, squeegee preparation, hardness categories of squeegee

blades, Variety of blade shape and application. Stencils-types, preparation methods of – knife cut stencils, photo stencils – Indirect stencil systems, Direct photo stencil systems, capillary systems, Direct/Indirect photo stencil systems. Explain about substrates and inks used in screen printing and various ink using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, stencils-classifications and preparation. 3 understanding content: history of screening printing, advantages, disadvantages and applications of screen printing. screen printing materials: screen, frame, squeegee and stencil. screens – multifilament, mono filaments, selecting mesh material, stretching screen fabric to frame, screen preparation, screen reclamation – trouble shooting clogged screens. care and storage of screens. the squeegee, squeegee considerations, squeegee preparation, hardness categories of squeegee blades, variety of blade shape and application. stencils-types, preparation methods of – knife cut stencils, photo stencils – indirect stencil systems, direct photo stencil systems, capillary systems, direct/indirect photo stencil systems. explain about substrates and inks used in screen printing and various ink should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Stencils-classifications and preparation. 3 Understanding Content: History of Screening Printing, Advantages, Disadvantages and Applications of screen printing. Screen printing materials: Screen, Frame, Squeegee and Stencil. Screens – multifilament, mono filaments, Selecting mesh material, stretching screen fabric to frame, screen preparation, screen reclamation – Trouble shooting clogged screens. Care and storage of screens. The squeegee, Squeegee considerations, squeegee preparation, hardness categories of squeegee blades, Variety of blade shape and application. Stencils-types, preparation methods of – knife cut stencils, photo stencils – Indirect stencil systems, Direct photo stencil systems, capillary systems, Direct/Indirect photo stencil systems. Explain about substrates and inks used in screen printing and various ink means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-സ്ക്രീൻ സ്റ്റൈംപ്സ്-കാർട്ടിഫിക്കേഷൻ and preparation. 3

Understanding Content: History of Screening Printing, Advantages, Disadvantages and Applications of screen printing. Screen printing materials: Screen, Frame, Squeegee and Stencil. Screens – multifilament, mono filaments, Selecting mesh material, stretching screen fabric to frame, screen preparation, screen reclamation – Trouble shooting clogged screens. Care and storage of screens. The squeegee, Squeegee considerations, squeegee preparation, hardness categories of squeegee blades, Variety of blade shape and application. Stencils-types, preparation methods of – knife cut stencils, photo stencils – Indirect stencil systems, Direct photo stencil systems, capillary systems, Direct/Indirect photo stencil systems. Explain about substrates and inks used in screen printing and various ink സ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ് definition/meaning സ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്. സ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്, steps, application സ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്. Technical terms English-സ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

and various materials required to achieve the screen printing process.

M1.01 History of Screening Printing, 1
Understanding

Merits, Demerits and applications of screen

M1.02 1
Understanding

printing.

Screen printing materials- Screen, Frame,

M1.03 3
Understanding

Squeegee and Stencil.

Screens – types, selection, stretching,

M1.04 3
Understanding

preparation & reclamation.

Trouble shooting clogged screens, Care and

M1.05 1
Understanding

storage of screens.

M1.06 3
Understanding

Squeegee considerations.

M1.07 3
Understanding

Stencils-classifications and preparation.

Content:

History of Screening Printing, Advantages, Disadvantages and Applications of
screen

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

drying methods and related facilities.

This module is presented from the official syllabus scope for Screen Printing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: drying methods and related facilities.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Screen-printing substrates

Screen-printing substrates is an official topic in Module 2 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Screen-printing substrates using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, screen-printing substrates should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Screen-printing substrates means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എസ്ക്രീൻ പ്രിന്റിംഗ് സബ്സ്ട്രേറ്റ്സ് എസ്ക്രീൻ പ്രിന്റിംഗ് സബ്സ്ട്രേറ്റ്സ്
definition/meaning എസ്ക്രീൻ പ്രിന്റിംഗ് സബ്സ്ട്രേറ്റ്സ്. എസ്ക്രീൻ പ്രിന്റിംഗ് സബ്സ്ട്രേറ്റ്സ്, steps, application
എസ്ക്രീൻ പ്രിന്റിംഗ് സബ്സ്ട്രേറ്റ്സ്. Technical terms English-എസ്ക്രീൻ പ്രിന്റിംഗ് സബ്സ്ട്രേറ്റ്സ്.

Screen-printing inks

Screen-printing inks is an official topic in Module 2 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Screen-printing inks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, screen-printing inks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Screen-printing inks means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എസ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ് എസ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്
definition/meaning എസ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്. എസ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്, steps, application
എസ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്. Technical terms English-എസ്ക്രീൻ പ്രിന്റിംഗ് ഇൻക്സ്.

Drying methods

Drying methods is an official topic in Module 2 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Drying methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, drying methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Drying methods means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Drying Equipments 3 Understanding Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process. Explain about on contact & off contact printing, specialized areas, image

Drying Equipments 3 Understanding Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process. Explain about on contact & off contact printing, specialized areas, image is an official topic in Module 2 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Drying Equipments 3 Understanding Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process. Explain about on contact & off contact printing, specialized areas, image using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, drying equipments 3 understanding screen-printing substrates: features & properties of-paper and paper board, wood, textiles, plastics, metals,

ceramics and glass. screen printing ink-solvent based, water based, radiation curable. drying methods - evaporation, oxidation, penetration, and polymerization. drying equipment - drying racks, wicket dryers, jet dryers, infrared dryers, ultraviolet dryers. flocking process. explain about on contact & off contact printing, specialized areas, image should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Drying Equipments 3 Understanding Screen-printing substrates: Features & properties of- Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink- Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process. Explain about on contact & off contact printing, specialized areas, image means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-☐☐ Drying Equipments 3 Understanding Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process. Explain about on contact & off contact printing, specialized areas, image ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

drying methods and related facilities.

M2.01	Screen-printing substrates	4
Understanding		
M2.02	Screen-printing inks	3
Understanding		
M2.03	Drying methods	4
Understanding		
M2.04	Drying Equipments	3
Understanding		
	Series test I	1

Contents:

Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process.

Explain about on contact & off contact printing, specialized areas,
image

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

transfer and registration.

This module is presented from the official syllabus scope for Screen Printing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: transfer and registration.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

On contact printing, Off contact printing. Understanding

On contact printing, Off contact printing. Understanding is an official topic in Module 3 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify On contact printing, Off contact printing. Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, on contact printing, off contact printing. understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What On contact printing, Off contact printing. Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓൺ കൺ ടാക്ട് പ്രിന്റിംഗ്, ഓഫ് കൺ ടാക്ട് പ്രിന്റിംഗ്.

Understanding ഓൺ കൺ ടാക്ട് പ്രിന്റിംഗ് definition/meaning ഓൺ കൺ ടാക്ട് പ്രിന്റിംഗ്. ഓൺ കൺ ടാക്ട് പ്രിന്റിംഗ് steps, application ഓൺ കൺ ടാക്ട് പ്രിന്റിംഗ്. Technical terms English-ഓൺ കൺ ടാക്ട് പ്രിന്റിംഗ്.

Specialized Areas of screen printing. 3 Understanding Image Transfer and Basic registration

Specialized Areas of screen printing. 3 Understanding Image Transfer and Basic registration is an official topic in Module 3 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Specialized Areas of screen printing. 3 Understanding Image Transfer and Basic registration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, specialized areas of screen printing. 3 understanding image transfer and basic registration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Specialized Areas of screen printing. 3 Understanding Image Transfer and Basic registration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്രത്യേകീകൃത സ്ക്രീൻ പ്രിന്റിംഗിന്റെ പ്രത്യേകതകൾ. 3 Understanding Image Transfer and Basic registration പ്രതിമ മാറ്റം പ്രതിമ മാറ്റം definition/meaning പ്രതിമ മാറ്റം പ്രതിമ മാറ്റം, steps, application പ്രതിമ മാറ്റം പ്രതിമ മാറ്റം പ്രതിമ മാറ്റം. Technical terms English-പ്രതിമ മാറ്റം പ്രതിമ മാറ്റം.

Understanding techniques. Method of halftone preparation for screen

Understanding techniques. Method of halftone preparation for screen is an official topic in Module 3 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding techniques. Method of halftone preparation for screen using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding techniques. method of halftone preparation for screen should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding techniques. Method of halftone preparation for screen means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- Understanding techniques. Method of halftone preparation for screen ഫ്ലാഷ് definition/meaning ഫ്ലാഷ്. ഫ്ലാഷ് ഫ്ലാഷ് ഫ്ലാഷ്, steps, application ഫ്ലാഷ് ഫ്ലാഷ് ഫ്ലാഷ്. Technical terms English- ഫ്ലാഷ് ഫ്ലാഷ് ഫ്ലാഷ്.

4 Understanding printing. On contact printing, Off contact printing. Specialized Areas - Printed circuit boards of screen printing. Image Transfer and Basic registration techniques. Method of halftone preparation for screen printing. Describe about various printing press classifications & configurations of

4 Understanding printing. On contact printing, Off contact printing. Specialized Areas - Printed circuit boards of screen printing. Image Transfer and Basic registration techniques. Method of halftone preparation for screen printing. Describe about various printing press classifications & configurations of is an official topic in Module 3 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding printing. On contact printing, Off contact printing. Specialized Areas - Printed circuit boards of screen printing. Image Transfer and Basic registration techniques. Method of halftone preparation for screen printing. Describe about various printing press classifications & configurations of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding printing. on contact printing, off contact printing. specialized areas - printed circuit boards of screen printing. image transfer and basic registration techniques. method of halftone preparation for screen printing. describe about various printing press classifications & configurations of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

transfer and registration.

M3.01 On contact printing, Off contact printing.

Understanding

M3.02 Specialized Areas of screen printing. 3

Understanding

Image Transfer and Basic registration

M3.03

Understanding

techniques.

Method of halftone preparation for screen

M3.04 4

Understanding

printing.

Contents:

On contact printing, Off contact printing. Specialized Areas - Printed circuit boards of

screen printing. Image Transfer and Basic registration techniques. Method of halftone

preparation for screen printing.

Describe about various printing press classifications & configurations of

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

screen-printing.

This module is presented from the official syllabus scope for Screen Printing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: screen-printing.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Manual Printing Process

Manual Printing Process is an official topic in Module 4 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Manual Printing Process using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, manual printing process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Manual Printing Process means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നു Manual Printing Process എന്നു definition/meaning എന്നു. എന്നു എന്നു, steps, application എന്നു എന്നു എന്നു. Technical terms English-എന്നു എന്നു എന്നു.

Semi automatic Screen Printing m/c.

Semi automatic Screen Printing m/c. is an official topic in Module 4 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Semi automatic Screen Printing m/c. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, semi automatic screen printing m/c. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Semi automatic Screen Printing m/c. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Semi automatic Screen Printing m/c.

ৱিকিপিডিয়াৰ পৰা definition/meaning ৰচনা কৰা হৈছে। ৱিকিপিডিয়াৰ পৰা steps, application ৰচনা কৰা হৈছে। Technical terms English-ৰ ৰচনা কৰা হৈছে।

Automatic Screen Printing m/c

Automatic Screen Printing m/c is an official topic in Module 4 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Automatic Screen Printing m/c using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, automatic screen printing m/c should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Automatic Screen Printing m/c means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓട്ടോമേറ്റഡ് സ്ക്രീൻ പ്രിന്റിംഗ് മ/c ഓട്ടോമേറ്റഡ് സ്ക്രീൻ പ്രിന്റിംഗ്
ഓട്ടോമേറ്റഡ് definition/meaning ഓട്ടോമേറ്റഡ് സ്ക്രീൻ പ്രിന്റിംഗ്. ഓട്ടോമേറ്റഡ് സ്ക്രീൻ പ്രിന്റിംഗ്, steps,
application ഓട്ടോമേറ്റഡ് സ്ക്രീൻ പ്രിന്റിംഗ്. Technical terms English-ഓട്ടോമേറ്റഡ്
ഓട്ടോമേറ്റഡ് സ്ക്രീൻ പ്രിന്റിംഗ്.

Screen Printing machines Configurations 3 Understanding Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing m/c. Screen Printing machines Configurations – Flat bet hinged frame, Flat bed vertical lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel m/c. Special Machine configurations. Text / Reference: T/R Book Title/Author Adams J.M, Faux,D.D and Rieber L.J, Printing Technology,5th Edition, T1 Delmar Publishers, NewYork. T2 P. Kipphan, Handbook of Print Media, Springer, 2002. R1 Babette Magee- Screen Printing Preview. R2 John Stephens.- Screen Printing. Online resources: SI. No Website Link 1 https://en.wikipedia.org/wiki/Screen_printing 2 <https://offbeatpress.com/basics-manual-screen-printing/> 3 http://printwiki.org/Screen_Printing 4 <https://static1.squarespace.com/static/ScreenprintManual.pdf> 5 <http://www.spmachine.ru/shelkografiya-en.htm>

Screen Printing machines Configurations 3 Understanding Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing m/c. Screen Printing machines Configurations – Flat bet hinged frame, Flat bed vertical lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel m/c. Special Machine configurations. Text / Reference: T/R Book Title/Author Adams J.M, Faux,D.D and Rieber L.J, Printing Technology,5th Edition, T1 Delmar Publishers, NewYork. T2 P. Kipphan, Handbook of Print Media, Springer, 2002. R1 Babette Magee- Screen Printing Preview. R2 John Stephens.- Screen Printing. Online resources: SI. No Website Link 1 https://en.wikipedia.org/wiki/Screen_printing 2 <https://offbeatpress.com/basics-manual-screen-printing/> 3 http://printwiki.org/Screen_Printing 4 <https://static1.squarespace.com/static/ScreenprintManual.pdf> 5 <http://www.spmachine.ru/shelkografiya-en.htm> is an official topic in Module 4 of Screen Printing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Screen Printing machines Configurations 3 Understanding Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing m/c. Screen Printing machines Configurations – Flat bet hinged frame, Flat

bed vertical lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel m/c. Special Machine configurations. Text / Reference: T/R Book Title/Author Adams J.M, Faux,D.D and Rieber L.J, Printing Technology,5th Edition, T1 Delmar Publishers, NewYork. T2 P. Kipphan, Handbook of Print Media, Springer, 2002. R1 Babette Magee- Screen Printing Preview. R2 John Stephens.- Screen Printing. Online resources: Sl. No Website Link 1 https://en.wikipedia.org/wiki/Screen_printing 2 <https://offbeatpress.com/basics-manual-screen-printing/> 3 http://printwiki.org/Screen_Printing 4 <https://static1.squarespace.com/static/ScreenprintManual.pdf> 5 <http://www.spmachine.ru/shelkografiya-en.htm> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, screen printing machines configurations 3 understanding manual printing process, semi automatic screen printing m/c. automatic screen printing m/c. screen printing machines configurations – flat bet hinged frame, flat bed vertical lift, cylinder-bed presses, container printing m/c, rotary screen printing m/c, carousel m/c. special machine configurations. text / reference: t/r book title/author adams j.m, faux,d.d and rieber l.j, printing technology,5th edition, t1 delmar publishers, newyork. t2 p. kipphan, handbook of print media, springer, 2002. r1 babette magee- screen printing preview. r2 john stephens.- screen printing. online resources: sl. no website link 1 https://en.wikipedia.org/wiki/screen_printing 2 <https://offbeatpress.com/basics-manual-screen-printing/> 3 http://printwiki.org/screen_printing 4 <https://static1.squarespace.com/static/screenprintmanual.pdf> 5 <http://www.spmachine.ru/shelkografiya-en.htm> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Screen Printing machines Configurations 3 Understanding Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing m/c. Screen Printing machines Configurations – Flat bet hinged frame, Flat bed vertical lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel m/c. Special Machine configurations. Text / Reference: T/R Book Title/Author Adams J.M, Faux,D.D and Rieber L.J, Printing Technology,5th Edition, T1 Delmar Publishers, NewYork. T2 P. Kipphan, Handbook of Print Media, Springer, 2002. R1 Babette Magee- Screen Printing Preview. R2 John Stephens.- Screen Printing. Online resources: Sl. No Website Link 1 https://en.wikipedia.org/wiki/Screen_printing 2 https://offbeatpress.com/basics-manual-screen-printing/ 3 http://printwiki.org/Screen_Printing 4 https://static1.squarespace.com/static/ScreenprintManual.pdf 5 http://www.spmachine.ru/shelkografiya-en.htm means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Screen Printing machines Configurations 3 Understanding Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing m/c. Screen Printing machines Configurations – Flat bed hinged frame, Flat bed vertical lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel m/c. Special Machine configurations. Text / Reference: T/R Book Title/Author Adams J.M, Faux,D.D and Rieber L.J, Printing Technology,5th Edition, T1 Delmar Publishers, NewYork. T2 P. Kipphan, Handbook of Print Media, Springer, 2002. R1 Babette Magee- Screen Printing Preview. R2 John Stephens.- Screen Printing. Online resources: Sl. No Website Link 1 https://en.wikipedia.org/wiki/Screen_printing 2 <https://offbeatpress.com/basics-manual-screen-printing/> 3 http://printwiki.org/Screen_Printing 4 <https://static1.squarespace.com/static/ScreenprintManual.pdf> 5 <http://www.spmachine.ru/shelkografiya-en.htm> □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

screen-printing.

M4.01	Manual Printing Process	3
Understanding		

M4.02	Semi automatic Screen Printing m/c.	4
Understanding		

M4.03	Automatic Screen Printing m/c	4
Understanding		

M4.04	Screen Printing machines Configurations	3
Understanding		

	Series test II	1
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Contents:

Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing

m/c. Screen Printing machines Configurations – Flat bed hinged frame, Flat bed vertical

lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel

m/c. Special Machine configurations.

Text / Reference:

T/R	Book Title/Author
T1	Adams J.M, Faux,D.D and Rieber L.J, Printing Technology,5th Edition, Delmar Publishers, NewYork.
T2	P. Kipphan, Handbook of Print Media, Springer, 2002.
R1	Babette Magee- Screen Printing Preview.
R2	John Stephens.- Screen Printing.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain History of Screening Printing, 1 Understanding Merits, Demerits and applications of screen. (3 marks)

Answer: Begin with the standard definition or identity of *History of Screening Printing, 1 Understanding Merits, Demerits and applications of screen*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 1 Understanding printing. Screen printing materials- Screen, Frame, (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding printing. Screen printing materials- Screen, Frame, .* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding Squeegee and Stencil. Screens - types, selection, stretching, (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Squeegee and Stencil. Screens - types, selection, stretching, .* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding preparation & reclamation. Trouble shooting clogged screens, Care and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding preparation & reclamation. Trouble shooting clogged screens, Care and. .* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Screen-printing substrates. (3 marks)

Answer: Begin with the standard definition or identity of *Screen-printing substrates*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Screen-printing inks. (3 marks)

Answer: Begin with the standard definition or identity of *Screen-printing inks*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Drying methods. (3 marks)

Answer: Begin with the standard definition or identity of *Drying methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Drying Equipments 3 Understanding Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket

dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process.

Explain about on contact & off contact printing, specialized areas, image. (3 marks)

Answer: Begin with the standard definition or identity of *Drying Equipments* 3 *Understanding Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process. Explain about on contact & off contact printing, specialized areas, image.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain On contact printing, Off contact printing. Understanding. (3 marks)

Answer: Begin with the standard definition or identity of *On contact printing, Off contact printing. Understanding.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Specialized Areas of screen printing. 3 Understanding Image Transfer and Basic registration. (3 marks)

Answer: Begin with the standard definition or identity of *Specialized Areas of screen printing. 3 Understanding Image Transfer and Basic registration.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understanding techniques. Method of halftone preparation for screen. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding techniques. Method of halftone preparation for screen*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding printing. On contact printing, Off contact printing. Specialized Areas - Printed circuit boards of screen printing. Image Transfer and Basic registration techniques. Method of halftone preparation for screen printing. Describe about various printing press classifications & configurations of. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding printing. On contact printing, Off contact printing. Specialized Areas - Printed circuit boards of screen printing. Image Transfer and Basic registration techniques. Method of halftone preparation for screen printing. Describe about various printing press classifications & configurations of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Manual Printing Process. (3 marks)

Answer: Begin with the standard definition or identity of *Manual Printing Process*. State the governing principle, list the key components or stages, and close with one relevant

application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Semi automatic Screen Printing m/c.. (3 marks)

Answer: Begin with the standard definition or identity of *Semi automatic Screen Printing m/c.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Automatic Screen Printing m/c. (3 marks)

Answer: Begin with the standard definition or identity of *Automatic Screen Printing m/c.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Screen Printing machines Configurations 3 Understanding Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing m/c. Screen Printing machines Configurations - Flat bed hinged frame, Flat bed vertical lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel m/c. Special Machine configurations. Text / Reference: T/R Book Title/Author Adams J.M, Faux,D.D and Rieber L.J, Printing Technology,5th Edition, T1 Delmar Publishers, NewYork. T2 P. Kipphan, Handbook of Print Media, Springer, 2002. R1 Babette Magee- Screen Printing Preview. R2 John Stephens.- Screen Printing. Online resources: Sl. No Website Link 1 https://en.wikipedia.org/wiki/Screen_printing 2 <https://offbeatpress.com/basics-manual-screen-printing/> 3 http://printwiki.org/Screen_Printing 4

<https://static1.squarespace.com/static/ScreenprintManual.pdf> 5

<http://www.spmachine.ru/shelkografiya-en.htm>. (3 marks)

Answer: Begin with the standard definition or identity of *Screen Printing machines Configurations* 3 *Understanding Manual Printing Process*, *Semi automatic Screen Printing m/c*. *Automatic Screen Printing m/c*. *Screen Printing machines Configurations – Flat bed hinged frame*, *Flat bed vertical lift*, *Cylinder-bed presses*, *Container printing m/c*, *Rotary Screen Printing m/c*, *Carousel m/c*. *Special Machine configurations*. Text / Reference: T/R Book Title/Author Adams J.M, Faux,D.D and Rieber L.J, *Printing Technology*,5th Edition, T1 Delmar Publishers, NewYork. T2 P. Kipphan, *Handbook of Print Media*, Springer, 2002. R1 Babette Magee- *Screen Printing Preview*. R2 John Stephens.- *Screen Printing*. Online resources: Sl. No Website Link 1 https://en.wikipedia.org/wiki/Screen_printing 2 <https://offbeatpress.com/basics-manual-screen-printing/> 3 http://printwiki.org/Screen_Printing 4 <https://static1.squarespace.com/static/ScreenprintManual.pdf> 5 <http://www.spmachine.ru/shelkografiya-en.htm>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **History of Screening Printing**, **1 Understanding Merits, Demerits and applications of screen**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Screen-printing substrates**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **On contact printing, Off contact printing. Understanding**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Manual Printing Process**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link
- https://en.wikipedia.org/wiki/Screen_printing

- <https://offbeatpress.com/basics-manual-screen-printing/>
http://printwiki.org/Screen_Printing
- <https://static1.squarespace.com/static/ScreenprintManual.pdf>
<http://www.spmachine.ru/shelkografiya-en.htm>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5103C. Verify institutional updates against the current official curriculum.